

# Side Effects

## How Drug Causes Side Effects

- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- By blocking dopamine 2 receptors excessively in the mesocortical and mesolimbic dopamine pathways, especially at high doses, it can cause worsening of negative and cognitive symptoms (neuroleptic-induced deficit syndrome)
- Blocking muscarinic cholinergic receptors can cause dry mouth, blurred vision, urinary retention, constipation, and paralytic ileus
- Antihistaminic actions may cause sedation, weight gain
- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
- Mechanism of weight gain and any possible increased incidence of diabetes or dyslipidemia with conventional antipsychotics is unknown

## Notable Side Effects

- \* Neuroleptic-induced deficit syndrome

- \* Akathisia
- \* Priapism
- \* Drug-induced parkinsonism
- \* Tardive dyskinesia, tardive dystonia
- \* Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration
- \* Galactorrhea, amenorrhea
- Dizziness, sedation
- Dry mouth, constipation, urinary retention, blurred vision
- Decreased sweating, depression
- Sexual dysfunction
- Hypotension, tachycardia, syncope
- Weight gain



### **Life-Threatening or Dangerous Side Effects**

- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- Rare jaundice, agranulocytosis
- Rare seizures

- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis

## Weight Gain



- Occurs in significant minority

## Sedation



- Occurs in significant minority

## What to Do About Side Effects

- Wait
- Wait
- Wait
- For drug-induced parkinsonism, add an anticholinergic agent
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Reduce the dose
- For sedation, take at night
- Switch to an atypical antipsychotic

- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain

### **Best Augmenting Agents for Side Effects**

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT<sub>2A</sub> antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

## **Dosing And Use**

### **Usual Dosage Range**

- Oral: 1–20 mg/day maintenance
- Intramuscular: generally 1/3 to 1/2 the oral dose
- Decanoate for intramuscular or subcutaneous administration: 12.5–100 mg/2 weeks maintenance (see Fluphenazine Decanoate section after Pearls for dosing and use)

### **Dosage Forms**

- Tablet 1 mg, 2.5 mg scored, 5 mg scored, 10 mg scored
- Decanoate for long-acting intramuscular or subcutaneous administration 25 mg/mL
- Injection for acute intramuscular administration 2.5 mg/mL
- Elixir 2.5 mg/5 mL
- Concentrate 5 mg/mL

### **How to Dose**

- Oral: initial 0.5–10 mg/day in divided doses; maximum 40 mg/day
- Intramuscular (short-acting): initial 1.25 mg; 2.5–10 mg/day can be given in divided doses every 6–8 hours; maximum dose generally 10 mg/day



### **Dosing Tips –Oral**

- Patients receiving atypical antipsychotics may occasionally require a “top-up” of a conventional antipsychotic to control aggression or violent behavior
- Fluphenazine tablets 2.5 mg, 5 mg, and 10 mg contain tartrazine, which can cause allergic reactions, especially in patients sensitive to aspirin
- Oral solution should not be mixed with drinks containing caffeine, tannic acid (tea), or pectinates (apple juice)

- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly
- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine
- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- For treatment resistance, consider obtaining plasma drug levels to guide dosing above recommended levels
- Treatment should be suspended if absolute neutrophil count falls below  $1000/\text{mm}^3$

### **Overdose**

- Drug-induced parkinsonism, coma, hypotension, sedation, seizures, respiratory depression

### **Long-Term Use**

- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

### **Habit Forming**

- No

## **How to Stop**

- Slow down-titration of oral formulation (over 6 to 8 weeks), especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)
- Rapid oral discontinuation may lead to rebound psychosis and worsening of symptoms
- If antiparkinson agents are being used, they should be continued for a few weeks after fluphenazine is discontinued

## **Pharmacokinetics**

- Mean half-life of oral formulation approximately 15 hours
- Mean half-life of intramuscular formulation approximately 6.8–9.6 days



## **Drug Interactions**

- May decrease the effects of levodopa, dopamine agonists
- May increase the effects of antihypertensive drugs except for guanethidine, whose antihypertensive actions fluphenazine may antagonize
- Additive effects may occur if used with CNS depressants
- Additive anticholinergic effects may occur if used with atropine or related compounds
- Alcohol and diuretics may increase the risk of hypotension

- Some patients taking a neuroleptic and lithium have developed an encephalopathic syndrome similar to neuroleptic malignant syndrome
- Combined use with epinephrine may lower blood pressure

### **Other Warnings/Precautions**

- If signs of neuroleptic malignant syndrome develop, treatment should be immediately discontinued
- Use cautiously in patients with alcohol withdrawal or convulsive disorders because of possible lowering of seizure threshold
- Avoid undue exposure to sunlight
- Use cautiously in patients with respiratory disorders
- Avoid extreme heat exposure
- Antiemetic effect can mask signs of other disorders or overdose
- Do not use epinephrine in event of overdose as interaction with some pressor agents may lower blood pressure
- Use only with caution if at all in Parkinson's disease or Lewy body dementia
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic

treatment, and recurrently for patients on long-term antipsychotic therapy

- As with any antipsychotic, use with caution in patients with history of seizures

### **Do Not Use**

- If patient is in a comatose state or has CNS depression
- If patient is taking cabergoline, pergolide, or metrizamide
- If there is a proven allergy to fluphenazine
- If there is a known sensitivity to any phenothiazine
- Decanoate and enanthate injectable formulations are contraindicated in children under age 12

## **Special Populations**

### **Renal Impairment**

- Use with caution; titration should be slower

### **Hepatic Impairment**

- Use with caution; titration should be slower

### **Cardiac Impairment**

- Cardiovascular toxicity can occur, especially orthostatic hypotension

## **Elderly**

- Titration should be slower; lower initial dose (1–2.5 mg/day)
- Elderly patients may be more susceptible to adverse effects
- Although conventional antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events

## **Children and Adolescents**

- Safety and efficacy not established
- Decanoate and enanthate injectable formulations are contraindicated in children under age 12
- Generally consider second-line after atypical antipsychotics

## **Pregnancy**

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001

- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- Reports of drug-induced parkinsonism, jaundice, hyperreflexia, hyporeflexia in infants whose mothers took a phenothiazine during pregnancy
- Fluphenazine should only be used during pregnancy if clearly indicated
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- Atypical antipsychotics may be preferable to conventional antipsychotics or anticonvulsant mood stabilizers if treatment is required during pregnancy

### **Breast Feeding**

- Some drug is found in mother's breast milk
- Effects on infant have been observed (dystonia, tardive dyskinesia, sedation)
- \* Recommended either to discontinue drug or bottle feed

## **The Art Of Psychopharmacology**

### **Potential Advantages**

- Intramuscular formulation for emergency use
- Relatively rapid onset of LAI (but see fluphenazine decanoate after Pearls section)

### **Potential Disadvantages**

- Patients with tardive dyskinesia
- Children
- Elderly

### **Primary Target Symptoms**

- Positive symptoms of psychosis
- Motor and autonomic hyperactivity
- Violent or aggressive behavior

### **Pearls**

- Fluphenazine is a high-potency phenothiazine
- Less risk of sedation and orthostatic hypotension but greater risk of drug-induced parkinsonism than with low-potency phenothiazines
- Not shown to be effective for behavioral problems in mental retardation
- Patients have very similar antipsychotic responses to any conventional antipsychotic, which is different from atypical

antipsychotics where antipsychotic responses of individual patients can occasionally vary greatly from one atypical antipsychotic to another

- Patients with inadequate responses to atypical antipsychotics may benefit from a trial of augmentation with a conventional antipsychotic such as fluphenazine or from switching to a conventional antipsychotic such as fluphenazine
- However, long-term polypharmacy with a combination of a conventional antipsychotic such as fluphenazine with an atypical antipsychotic may combine their side effects without clearly augmenting the efficacy of either
- For treatment-resistant patients, especially those with impulsivity, aggression, violence, and self-harm, long-term polypharmacy with 2 atypical antipsychotics or with 1 atypical antipsychotic and 1 conventional antipsychotic may be useful or even necessary while closely monitoring
- In such cases, it may be beneficial to combine 1 depot antipsychotic with 1 oral antipsychotic
- Although a frequent practice by some prescribers, adding 2 conventional antipsychotics together has little rationale and may reduce tolerability without clearly enhancing efficacy

### **Fluphenazine Decanoate**

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Vehicle

Sesame Oil

|                                       |                               |
|---------------------------------------|-------------------------------|
| T <sub>max</sub>                      | 0.3–1.5 days                  |
| T <sub>1/2</sub> with multiple dosing | 14 days                       |
| Able to be loaded                     | Yes                           |
| Time to reach steady state            | 4–6 weeks with loading        |
| Dosing schedule (maintenance)         | 2 weeks                       |
| Injection site                        | Intramuscular or subcutaneous |
| Needle gauge                          | 21                            |
| Dosage forms                          | 25 mg                         |
| Injection volume                      | 25 mg/mL                      |

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### **Usual Dosage Range**

- 12.5–100 mg/2 weeks maintenance

### **How to Dose**

- Conversion from oral: can either supplement with oral formulation at half-dose for at least 2 weeks OR use weekly loading injections of 1.6 times the oral daily dose (mg/day) for 4–6 weeks

### **Dosing Tips**

- With LAIs, the absorption rate constant is slower than the elimination rate constant, thus resulting in “flip-flop” kinetics –i.e.,

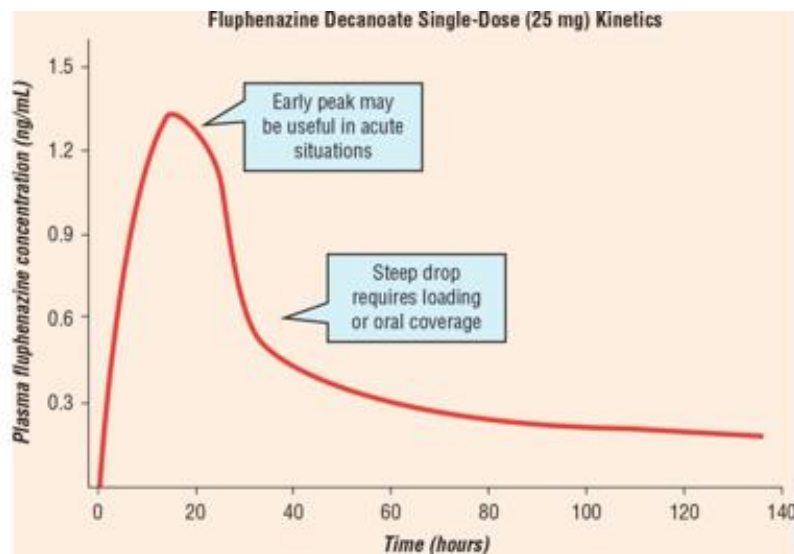
time to steady state is a function of absorption rate, while concentration at steady state is a function of elimination rate

- In general, 5 half-lives of any medication are needed to achieve 97% of steady-state levels
- The long half-lives of depot antipsychotics mean that one must either adequately load the dose (if possible) or provide oral supplementation
- The failure to adequately load the dose leads either to prolonged cross-titration from oral antipsychotic or to sub-therapeutic antipsychotic plasma levels for weeks or months in patients who are not receiving (or adhering to) oral supplementation
- Because plasma antipsychotic levels increase gradually over time, dose requirements may ultimately decrease from initial; obtaining periodic plasma levels can be beneficial to prevent unnecessary plasma level creep
- The time to get a blood level for patients receiving LAI is the morning of the day they will receive their next injection
- Advantages: early peak (see following graph) may be beneficial in the management of acute patients
- Disadvantages: early peak also carries risk of drug-induced parkinsonism or akathisia in first 48 hours; 2-week injection schedule; higher incidence of local site reactions (due to sesame oil vehicle)

- Response threshold is 0.81 ng/mL; plasma levels greater than 2–3 ng/mL are generally not well tolerated, although treatment-resistant cases might tolerate and respond to plasma levels up to 4 ng/mL, which is the point of futility if no response and even if tolerated

## The Art Of Switching

🔄 Switching from Oral Antipsychotics to Fluphenazine Decanoate



- Discontinuation of oral antipsychotic can begin immediately if adequate loading is pursued; otherwise, supplement with oral formulation at half-dose for at least 2 weeks
- How to discontinue oral formulations
- Down-titration is not required for: amisulpride, aripiprazole, brexpiprazole, cariprazine, paliperidone ER
- 1-week down-titration is required for: iloperidone, lurasidone, risperidone, ziprasidone

- 3–4-week down-titration is required for: asenapine, olanzapine, quetiapine
- 4+-week down-titration is required for: clozapine
- For patients taking benzodiazepine or anticholinergic medication, this can be continued during cross-titration to help alleviate side effects such as insomnia, agitation, and/or psychosis. Once the patient is stable on LAI, these can be tapered one at a time as appropriate.

### *Suggested Reading*

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Pharmacokinetic considerations for antipsychotic drugs . *Clin*  
*Pharmacokinet* 1995 ;**28** (6 ):494–504 .

Flurazepam

## Therapeutics

### Brands

- Dalmane
- *see index for additional brand names*

### Generic?

Yes

### Class

- Neuroscience-based Nomenclature: GABA positive allosteric modulator (GABA-PAM)
- Benzodiazepine (hypnotic)

### Commonly Prescribed for

*(bold for FDA approved)*

- **Insomnia characterized by difficulty in falling asleep, frequent nocturnal awakenings, and/or early morning awakening**
- **Recurring insomnia or poor sleeping habits**
- **Acute or chronic medical situations requiring restful sleep**
- Catatonia

## **How the Drug Works**

- Binds to benzodiazepine receptors at the GABA-A ligand-gated chloride channel complex
- Enhances the inhibitory effects of GABA
- Boosts chloride conductance through GABA-regulated channels
- Inhibitory actions in sleep centers may provide sedative hypnotic effects

## **How Long Until It Works**

- Generally takes effect in less than an hour

## **If It Works**

- Improves quality of sleep
- Effects on total wake-time and number of nighttime awakenings may be decreased over time

## **If It Doesn't Work**

- If insomnia does not improve after 7–10 days, it may be a manifestation of a primary psychiatric or physical illness such as obstructive sleep apnea or restless leg syndrome, which requires independent evaluation
- Increase the dose
- Improve sleep hygiene

- Switch to another agent

### **Best Augmenting Combos for Partial Response or Treatment Resistance**

- Generally, best to switch to another agent
- Trazodone
- Agents with antihistamine actions (e.g., diphenhydramine, TCAs)

### **Tests**

- In patients with seizure disorders, concomitant medical illness, and/or those with multiple concomitant long-term medications, periodic liver tests and blood counts may be prudent

## **Side Effects**

### **How Drug Causes Side Effects**

- Same mechanism for side effects as for therapeutic effects –namely due to excessive actions at benzodiazepine receptors
- Actions at benzodiazepine receptors that carry over to the next day can cause daytime sedation, amnesia, and ataxia
- Long-term adaptations in benzodiazepine receptors may explain the development of dependence, tolerance, and withdrawal

### **Notable Side Effects**

- \* Sedation, fatigue, depression
- \* Dizziness, ataxia, slurred speech, weakness
- \* Forgetfulness, confusion
- \* Hyperexcitability, nervousness
- Rare hallucinations, mania
- Rare hypotension
- Hypersalivation, dry mouth
- Rebound insomnia when withdrawing from long-term treatment



### **Life-Threatening or Dangerous Side Effects**

- Respiratory depression, especially when taken with CNS depressants in overdose
- Rare hepatic dysfunction, renal dysfunction, blood dyscrasias

## **Weight Gain**



- Reported but not expected

## **Sedation**



- Many experience and/or can be significant in amount

## **What to Do About Side Effects**

- Wait
- To avoid problems with memory, only take flurazepam if planning to have a full night's sleep
- Lower the dose
- Switch to a shorter-acting sedative hypnotic
- Switch to a non-benzodiazepine hypnotic
- Administer flumazenil if side effects are severe or life-threatening

### **Best Augmenting Agents for Side Effects**

- Many side effects cannot be improved with an augmenting agent

## **Dosing And Use**

### **Usual Dosage Range**

- 15–30 mg/day at bedtime for 7–10 days

### **Dosage Forms**

- Capsule 15 mg, 30 mg

### **How to Dose**

- 15 mg/day at bedtime; may increase to 30 mg/day at bedtime if ineffective



## **Dosing Tips**

- \* Because flurazepam tends to accumulate over time, perhaps not the best hypnotic for chronic nightly use
- Use lowest possible effective dose and assess need for continued treatment regularly
- Flurazepam should generally not be prescribed in quantities greater than a 1-month supply
- Patients with lower body weights may require lower doses
- Risk of dependence may increase with dose and duration of treatment

## **Overdose**

- No death reported in monotherapy; sedation, slurred speech, poor coordination, confusion, coma, respiratory depression

## **Long-Term Use**

- Not generally intended for long-term use
- \* Because of its relatively longer half-life, flurazepam may cause some daytime sedation and/or impaired motor/cognitive function, and may do so progressively over time

## **Habit Forming**

- Flurazepam is a Schedule IV drug

- Some patients may develop dependence and/or tolerance; risk may be greater with higher doses
- History of drug addiction may increase risk of dependence

### **How to Stop**

- If taken for more than a few weeks, taper to reduce chances of withdrawal effects
- Patients with seizure history may seize upon sudden withdrawal
- Rebound insomnia may occur the first 1–2 nights after stopping
- For patients with severe problems discontinuing a benzodiazepine, dosing may need to be tapered over many months (i.e., reduce dose by 1% every 3 days by crushing tablet and suspending or dissolving in 100 mL of fruit juice and then disposing of 1 mL and drinking the rest; 3–7 days later, dispose of 2 mL, and so on). This is both a form of very slow biological tapering and a form of behavioral desensitization

### **Pharmacokinetics**

- Elimination half-life approximately 24–100 hours
- Active metabolites



### **Drug Interactions**

- Cimetidine may decrease flurazepam clearance and thus raise flurazepam levels